**Dominic Bumpus Week 6 R Assignment Code Output**

```
> # Load Libraries
> library(caret)
Loading required package: ggplot2
Loading required package: lattice> library(MASS)
> library(randomForest)
randomForest 4.7-1.2
Type rfNews() to see new features/changes/bug fixes.

Attaching package: ‘randomForest’

The following object is masked from ‘package:ggplot2’:

    margin> library(gbm)
Loaded gbm 2.2.3
This version of gbm is no longer under development. Consider transitioning to gbm3,
https://github.com/gbm-developers/gbm3> library(rpart)
> library(rpart.plot)
> library(pROC)
Type 'citation("pROC")' for a citation.

Attaching package: ‘pROC’

The following objects are masked from ‘package:stats’:

    cov, smooth, var>
> set.seed(12345)
```

```
>

>

> # Load Data

> hmeq <- read.csv("C:\\Users\\Wolfh\\Downloads\\HMEQ_Scrubbed
(2)\\HMEQ_Scrubbed.csv")

>

> # Ensure data types

> hmeq$TARGET_BAD_FLAG <- as.factor(hmeq$TARGET_BAD_FLAG)

>

> # Remove rows with missing target values

> hmeq <- hmeq[!is.na(hmeq$TARGET_BAD_FLAG), ]

>

>

> # TRAIN/TEST SPLIT

> train_index <- createDataPartition(hmeq$TARGET_BAD_FLAG, p = 0.7, list = FALSE)

> train <- hmeq[train_index, ]

> test  <- hmeq[-train_index, ]

>

>

> # Logistic Regression

> log_full <- glm(TARGET_BAD_FLAG ~ .,

+               data = train,

+               family = binomial)

Warning messages:

1: glm.fit: algorithm did not converge

2: glm.fit: fitted probabilities numerically 0 or 1 occurred >

>

> # Backward Selection
```

```
> log_backward <- stepAIC(log_full,
+                         direction = "backward",
+                         trace = FALSE)
There were 50 or more warnings (use warnings() to see the first 50)>
> # Forward Selection
> log_null <- glm(TARGET_BAD_FLAG ~ 1,
+                 data = train,
+                 family = binomial)
>
> log_forward <- stepAIC(log_null,
+                        scope = list(lower = log_null, upper = log_full),
+                        direction = "forward",
+                        trace = FALSE)
There were 50 or more warnings (use warnings() to see the first 50)>
> # Display Important Variables
> summary(log_backward)

Call:
glm(formula = TARGET_BAD_FLAG ~ TARGET_LOSS_AMT, family = binomial,
    data = train)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)     -20.5474   303.9614  -0.068    0.946
TARGET_LOSS_AMT   0.1478     1.9812   0.075    0.941

(Dispersion parameter for binomial family taken to be 1)
```

```
    Null deviance: 4.1719e+03  on 4172  degrees of freedom
Residual deviance: 1.5000e-05  on 4171  degrees of freedom
AIC: 4

Number of Fisher Scoring iterations: 25

> summary(log_forward)

Call:
glm(formula = TARGET_BAD_FLAG ~ TARGET_LOSS_AMT, family = binomial,
    data = train)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)     -20.5474   303.9614  -0.068    0.946
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    Null deviance: 4.1719e+03  on 4172  degrees of freedom
Residual deviance: 1.5000e-05  on 4171  degrees of freedom
AIC: 4

Number of Fisher Scoring iterations: 25

>
> # Random Forest (Classification)
> rf_class <- randomForest(TARGET_BAD_FLAG ~ .,
```

```
+                 data = train,
+                 ntree = 300,
+                 importance = TRUE)
>
> varImpPlot(rf_class)
>
> # Gradient Boosting
> train_gbm <- train
> train_gbm$TARGET_BAD_FLAG <-
as.numeric(as.character(train_gbm$TARGET_BAD_FLAG))
>
> gbm_class <- gbm(TARGET_BAD_FLAG ~ .,
+           data = train_gbm,
+           distribution = "bernoulli",
+           n.trees = 300,
+           interaction.depth = 3,
+           shrinkage = 0.05,
+           verbose = FALSE)
>
> # ROC Curves
> log_prob <- predict(log_backward, test, type = "response")
> rf_prob  <- predict(rf_class, test, type = "prob")[,2]
> gbm_prob <- predict(gbm_class, test, n.trees = 300, type = "response")
>
> roc_log <- roc(test$TARGET_BAD_FLAG, log_prob)
Setting levels: control = 0, case = 1
Setting direction: controls < cases> roc_rf  <- roc(test$TARGET_BAD_FLAG, rf_prob)
Setting levels: control = 0, case = 1
```

```
Setting direction: controls < cases> roc_gbm <- roc(test$TARGET_BAD_FLAG, gbm_prob)
Setting levels: control = 0, case = 1
Setting direction: controls < cases>
> plot(roc_log, col="blue", lwd=2, main="ROC Curves")
> plot(roc_rf, col="red", add=TRUE, lwd=2)
> plot(roc_gbm, col="green", add=TRUE, lwd=2)
>
> legend("bottomright",
+       legend=c("Logistic","Random Forest","GBM"),
+       col=c("blue","red","green"),
+       lwd=2)
>
> auc_log <- auc(roc_log)
> auc_rf  <- auc(roc_rf)
> auc_gbm <- auc(roc_gbm)
>
> print(auc_log)
Area under the curve: 1
> print(auc_rf)
Area under the curve: 1
> print(auc_gbm)
Area under the curve: 1
>
> # LINEAR REGRESSION
>
> # Use only defaults
> train_sev <- subset(train, TARGET_BAD_FLAG == 1)
> test_sev  <- subset(test, TARGET_BAD_FLAG == 1)
```

```
>
> train_sev$TARGET_BAD_FLAG <- NULL
> test_sev$TARGET_BAD_FLAG  <- NULL
>
> # Remove rows with missing loss
> train_sev <- train_sev[!is.na(train_sev$TARGET_LOSS_AMT), ]
> test_sev  <- test_sev[!is.na(test_sev$TARGET_LOSS_AMT), ]
>
>
> # Linear Regression (Full)
> lm_full <- lm(TARGET_LOSS_AMT ~ ., data=train_sev)
>
>
> # Backward Selection
> lm_backward <- stepAIC(lm_full,
+                        direction="backward",
+                        trace=FALSE)
>
>
> # Forward Selection
> lm_null <- lm(TARGET_LOSS_AMT ~ 1, data=train_sev)
>
> lm_forward <- stepAIC(lm_null,
+                       scope=list(lower=lm_null, upper=lm_full),
+                       direction="forward",
+                       trace=FALSE)
>
> summary(lm_backward)
```

```
Call:
lm(formula = TARGET_LOSS_AMT ~ LOAN + M_MORTDUE + IMP_VALUE +
   IMP_YOJ + IMP_DEROG + M_DEROG + IMP_DELINQ + IMP_CLAGE +
   M_CLAGE + IMP_CLNO + M_CLNO + IMP_DEBTINC + M_DEBTINC + FLAG.Job.Sales +
   FLAG.Job.Self + FLAG.Reason.DebtCon + FLAG.Reason.HomeImp,
   data = train_sev)

Residuals:
    Min      1Q   Median      3Q     Max
-18641.0  -1108.1    165.6   1656.7  14712.5

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)      -1.063e+04  9.069e+02 -11.719  < 2e-16 ***
LOAN              7.983e-01  1.299e-02  61.451  < 2e-16 ***
M_MORTDUE        -1.058e+03  4.721e+02  -2.241   0.0253 *
IMP_VALUE        -9.272e-03  2.023e-03  -4.584 5.27e-06 ***
IMP_YOJ          -9.492e+01  1.901e+01  -4.992 7.30e-07 ***
IMP_DEROG         3.663e+02  9.096e+01   4.028 6.16e-05 ***
M_DEROG           1.064e+03  7.344e+02   1.448   0.1479
IMP_DELINQ        7.268e+02  7.112e+01  10.218  < 2e-16 ***
IMP_CLAGE        -1.944e+01  1.630e+00 -11.930  < 2e-16 ***
M_CLAGE          -5.101e+03  9.060e+02  -5.631 2.47e-08 ***
IMP_CLNO          2.080e+02  1.253e+01  16.605  < 2e-16 ***
M_CLNO            5.474e+03  1.310e+03   4.178 3.26e-05 ***
IMP_DEBTINC       1.083e+02  1.282e+01   8.450  < 2e-16 ***
M_DEBTINC         5.542e+03  2.702e+02  20.509  < 2e-16 ***
```

```
FLAG.Job.Sales       1.535e+03  8.290e+02   1.851   0.0645 .
FLAG.Job.Self        2.433e+03  5.835e+02   4.170 3.37e-05 ***
FLAG.Reason.DebtCon  2.546e+03  6.507e+02   3.912 9.91e-05 ***
FLAG.Reason.HomeImp  9.538e+02  6.721e+02   1.419   0.1563
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3592 on 815 degrees of freedom
Multiple R-squared:  0.8958,  Adjusted R-squared:  0.8936
F-statistic: 411.9 on 17 and 815 DF,  p-value: < 2.2e-16

> summary(lm_forward)

Call:
lm(formula = TARGET_LOSS_AMT ~ LOAN + IMP_CLNO + M_DEBTINC +
    IMP_CLAGE + IMP_DELINQ + IMP_DEBTINC + FLAG.Reason.DebtCon +
    IMP_YOJ + IMP_VALUE + FLAG.Job.Self + IMP_DEROG + M_DEROG +
    M_CLAGE + M_CLNO + M_MORTDUE + FLAG.Job.Sales + FLAG.Reason.HomeImp,
    data = train_sev)

Residuals:
     Min       1Q   Median       3Q      Max
-18641.0  -1108.1    165.6   1656.7  14712.5

Coefficients:
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(Intercept)        -1.063e+04  9.069e+02 -11.719  < 2e-16 ***
LOAN                7.983e-01  1.299e-02  61.451  < 2e-16 ***
```

```
IMP_CLNO           2.080e+02  1.253e+01  16.605  < 2e-16 ***
M_DEBTINC          5.542e+03  2.702e+02  20.509  < 2e-16 ***
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FLAG.Job.Self      2.433e+03  5.835e+02   4.170 3.37e-05 ***
IMP_DEROG          3.663e+02  9.096e+01   4.028 6.16e-05 ***
M_DEROG            1.064e+03  7.344e+02   1.448   0.1479
M_CLAGE           -5.101e+03  9.060e+02  -5.631 2.47e-08 ***
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FLAG.Reason.HomeImp  9.538e+02  6.721e+02   1.419   0.1563
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Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3592 on 815 degrees of freedom
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F-statistic: 411.9 on 17 and 815 DF,  p-value: < 2.2e-16

>
>
> # RMSE Function
> rmse <- function(actual, predicted){
+   sqrt(mean((actual - predicted)^2))
```

```
+ }
>
> # Predictions
> pred_full <- predict(lm_full, test_sev)
> pred_back <- predict(lm_backward, test_sev)
> pred_forw <- predict(lm_forward, test_sev)
>
> rmse_full <- rmse(test_sev$TARGET_LOSS_AMT, pred_full)
> rmse_back <- rmse(test_sev$TARGET_LOSS_AMT, pred_back)
> rmse_forw <- rmse(test_sev$TARGET_LOSS_AMT, pred_forw)
>
> print(rmse_full)
[1] 3667.799
> print(rmse_back)
[1] 3617.712
> print(rmse_forw)
[1] 3617.712
>
>
> # PROBABILITY / SEVERITY MODEL
>
> # Probability of Default
> prob_default <- predict(log_backward, test, type="response")
> summary(log_backward)


Call:
glm(formula = TARGET_BAD_FLAG ~ TARGET_LOSS_AMT, family = binomial,
    data = train)
```

```
Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)     -20.5474   303.9614  -0.068    0.946
TARGET_LOSS_AMT   0.1478     1.9812   0.075    0.941

(Dispersion parameter for binomial family taken to be 1)

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AIC: 4

Number of Fisher Scoring iterations: 25

>
> # Severity Prediction (Linear model)
> sev_pred <- predict(lm_backward, test)
>
> sev_pred[is.na(sev_pred)] <- 0
> summary(log_backward)

Call:
glm(formula = TARGET_BAD_FLAG ~ TARGET_LOSS_AMT, family = binomial,
    data = train)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)     -20.5474   303.9614  -0.068    0.946
```

```
TARGET_LOSS_AMT   0.1478     1.9812   0.075    0.941

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 4.1719e+03  on 4172  degrees of freedom
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AIC: 4

Number of Fisher Scoring iterations: 25

>
> # Expected Loss
> expected_loss <- prob_default * sev_pred
>
> # RMSE for Probability/Severity Model
> rmse_ps <- rmse(test$TARGET_LOSS_AMT, expected_loss)
>
> print(rmse_ps)
[1] 1614.718
```
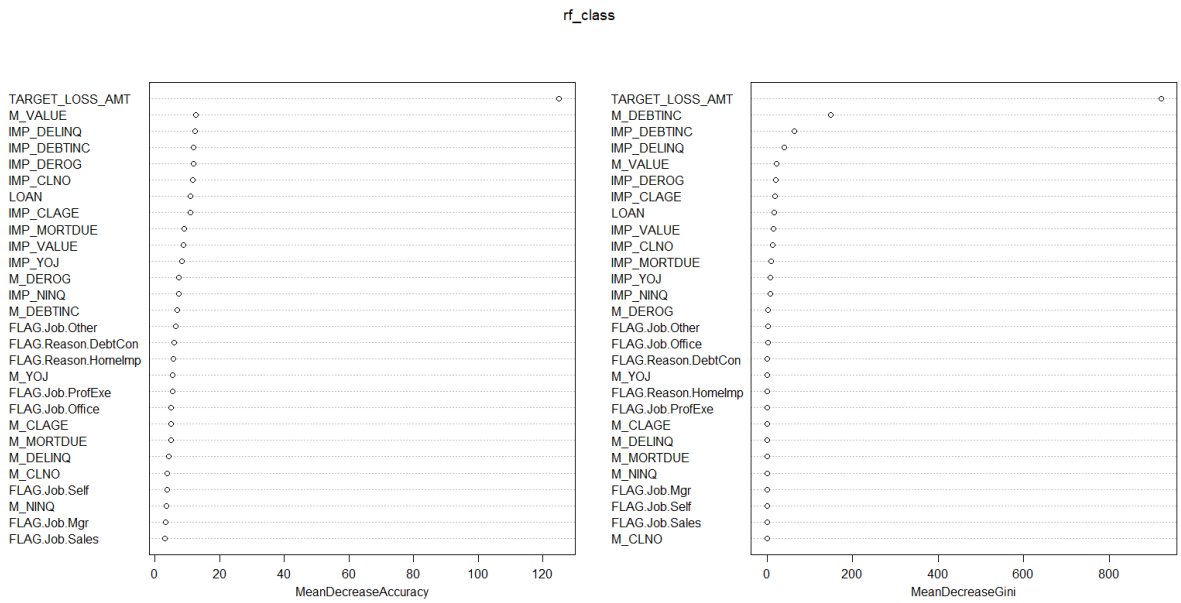
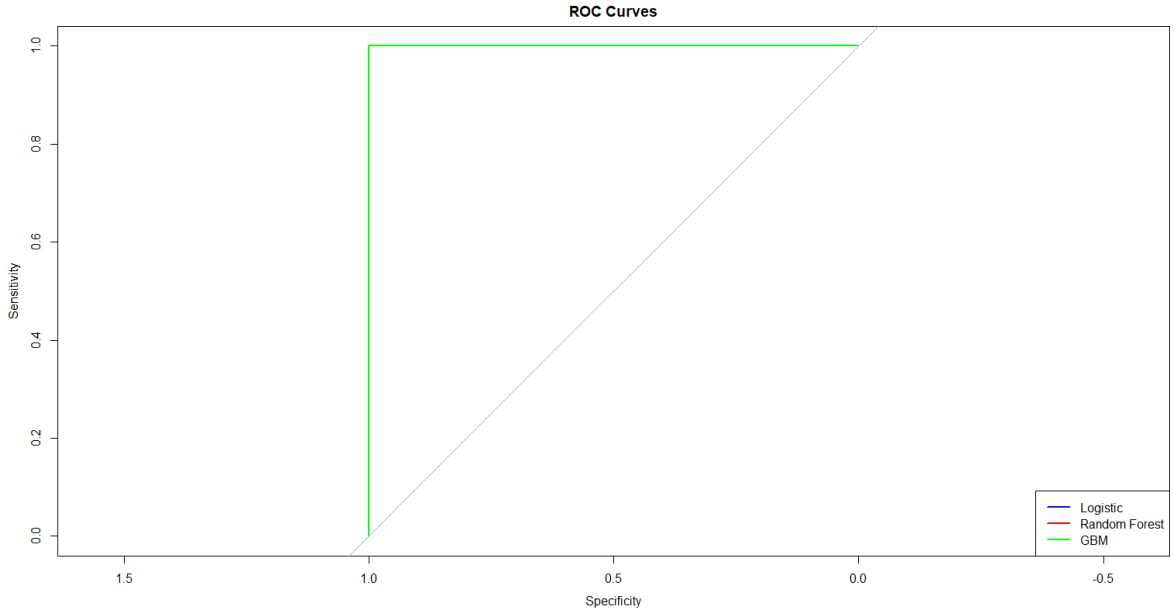

**Graphics**


rf_class
TARGET_LOSS_AMT
M_VALUE
IMP_DELINQ
IMP_DEBTINC
IMP_DEROG
IMP_CLNO
LOAN
IMP_CLAGE
IMP_MORTDUE
IMP_VALUE
IMP_YOJ
M_DEROG
IMP_NINQ
M_DEBTINC
FLAG.Job.Other
FLAG.Reason.DebtCon
FLAG.Reason.HomeImp
M_YOJ
FLAG.Job.ProfExe
FLAG.Job.Office
M_CLAGE
M_MORTDUE
M_DELINQ
M_CLNO
FLAG.Job.Self
M_NINQ
FLAG.Job.Mgr
FLAG.Job.Sales
0
20
40
60
80
100
120
MeanDecreaseAccuracy
TARGET_LOSS_AMT
M_DEBTINC
IMP_DEBTINC
IMP_DELINQ
M_VALUE
IMP_DEROG
IMP_CLAGE
LOAN
IMP_VALUE
IMP_CLNO
IMP_MORTDUE
IMP_YOJ
IMP_NINQ
M_DEROG
FLAG.Job.Other
FLAG.Job.Office
FLAG.Reason.DebtCon
M_YOJ
FLAG.Reason.HomeImp
FLAG.Job.ProfExe
M_CLAGE
M_DELINQ
M_MORTDUE
M_NINQ
FLAG.Job.Mgr
FLAG.Job.Self
FLAG.Job.Sales
M_CLNO
0
200
400
600
800
MeanDecreaseGini


ROC Curves
Sensitivity
0.0
0.2
0.4
0.6
0.8
1.0
1.5
1.0
0.5
0.0
-0.5
Specificity
Logistic
Random Forest
GBM